Mock for Quiz1

This assignment will not be graded and is only for practice.
If $A + 4I$ = 0A + 4I = 0, where AA is a $3 \times 3$3 × 3 matrix and II is an identity matrix of order 3, then find out the $det(A)$det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -64
1 point
2 points
If $\{v_1, v_2, v_3\}${v1, v2, v3} forms a basis of R^3 R3, then which of the following are true?

☐ $\{v_1, v_2, v_1 + v_3\}${v1, v2, v1 + v3} forms a basis of R^3 R3.

☐ $\{v_1, v_1 + v_2, v_1 + v_3\}${v1, v1 + v2, v1 + v3} forms a basis of R^3 R3.

☐ $\{v_1, v_1 + v_2, v_1 - v_3\}${v1, v1 + v2, v1 − v3} forms a basis of R^3 R3.

☐ $\{v_1, v_1 - v_2, v_1 - v_3\}${v1, v1 − v2, v1 − v3} forms a basis of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{v_1, v_2, v_1 + v_3\}${v1, v2, v1 + v3} forms a basis of R^3 R3.
$\{v_1, v_1 + v_2, v_1 + v_3\}${v1, v1 + v2, v1 + v3} forms a basis of R^3 R3.
$\{v_1, v_1 + v_2, v_1 - v_3\}${v1, v1 + v2, v1 − v3} forms a basis of R^3 R3.
$\{v_1, v_1 - v_2, v_1 - v_3\}${v1, v1 − v2, v1 − v3} forms a basis of R^3 R3.
2 points
Let the set $\{(a, b), (c, d)\}${(a, b), (c, d)} be a linearly independent subset of R^2 R2. Choose the set of correct options.

☐ $\{(a, b, 0), (c, d, 0)\}${(a, b, 0), (c, d, 0)} must be a linearly independent subset of R^3 R3.

☐ $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}${(a, b, 0), (c, d, 0), (0, 0, 1)} must be a linearly independent subset of R^3 R3.

☐ $\{(a, b, 0), (c, d, 0), (1, 0, 0)\}${(a, b, 0), (c, d, 0), (1, 0, 0)} must be a linearly independent subset of R^3 R3.

☐ $\{(a, b, 0), (c, d, 0), (0, 1, 0)\}${(a, b, 0), (c, d, 0), (0, 1, 0)} must be a linearly independent subset of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(a, b, 0), (c, d, 0)\}${(a, b, 0), (c, d, 0)} must be a linearly independent subset of R^3 R3.
$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}${(a, b, 0), (c, d, 0), (0, 0, 1)} must be a linearly independent subset of R^3 R3.

Suppose $A = \begin{bmatrix} -3 & 5 & -1 \\ 0 & -2 & 3 \\ 7 & -4 & 1 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 & x \\ x & -5 & 2 \\ -2 & 7 & 0 \end{bmatrix}$B = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$, where $x \in N$x ∈ N. If $det(A) = det(B)$det(A) = det(B), then find the value of xx.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

2 points

Consider the following system of linear equations:

$$\begin{aligned} x - 3y + 2z &= 5 - a \\ 3x - 4y + z &= a + 1 \\ -3x + 9y - 6z &= 2a \end{aligned}$$x − 3y + 2z = 5 − a 3x − 4y + z = a + 1x − 3y + 2z =3x − 4y + z =−3x + 9y − 6z = 5 − a a + 1 2a

where $a \in R$a ∈ R. If the given system of linear equations has infinitely many solutions, then what is the value of aa?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 15

2 points

2 points

Consider the set $S = \{(1, -1, 0), (0, 1, 1), (2, 0, 1)\}$S = {(1, −1, 0), (0, 1, 1), (2, 0, 1)}. Choose the correct set of options.

☐ $(7, -2, 2) \in$(7, −2, 2) ∈ Span(S)(S)

☐ Span(S)(S) is a proper subspace of R^3R3.

☐ Span(S)(S) =Span$\{(1, -1, 0), (0, 1, 1)\}${(1, −1, 0), (0, 1, 1)}

☐ dim(Span(S)(S))=33.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, -2, 2) \in$(7, −2, 2) ∈ Span(S)(S)

dim(Span(S)(S))=33.

Suppose VV is a vector space defined as $V = \{A | A \in M_{3\times 3}(R)$V = {A ∣ A ∈ M3×3(R), the sum of the diagonal entries of AA is zero }}. What is the dimension of VV ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

2 points

2 points

Consider the following sets with the given operations defined as follows:

$V_1 = \{(a, b) \mid a, b \in R\}$
Addition: $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2)$;
$(a_1, b_1), (a_2, b_2) \in V$ V1 =
Scalar multiplication: $\alpha(a, b) = (\alpha a, b)$;
$(a, b) \in V, \alpha \in R$

{(a, b) | a, b ∈ R}Addition: (a1, b1) + (a2, b2) = (a1 + a2, b1 + b2);
(a1, b1), (a2, b2) ∈ V Scalar multiplication: α(a, b) = (αa, b);
(a, b) ∈ V , α ∈ R

$V_2 = \{(a, b, c) \mid a, b, c \in R \text{and} ab + bc = 0\}$
Addition: $(a_1, b_1, c_1) + (a_2, b_2, c_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2)$;
$(a_1, b_1, c_1), (a_2, b_2, c_2) \in V$ V2 =
Scalar multiplication: $\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$;
$(a, b, c) \in V, \alpha \in R$

{(a, b, c) | a, b, c ∈ R and ab + bc = 0}Addition: (a1, b1, c1) + (a2, b2, c2) = (a1 + a2, b1 + b2, c1 + c2);
(a1, b1, c1), (a2, b2, c2) ∈ V Scalar multiplication: α(a, b, c) =
(αa, αb, αc); (a, b, c) ∈ V , α ∈ R

Choose the set of correct options.

- ☐ V_1 V1 is a vector space with respect the given operations.
- ☐ V_1 V1 is not a vector space with respect the given operations.
- ☐ V_2 V2 is a vector space with respect the given operations.
- ☐ V_2 V2 is not a vector space with respect the given operations.

No, the answer is incorrect.
Score: 0
Accepted Answers:
V_1 V1 is not a vector space with respect the given operations.
V_2 V2 is not a vector space with respect the given operations.
2 points
Find out the correct statements from the following:

- ☐ The set of all invertible matrices of order 22 forms a subspace of $M_2(R)$M2(R) with respect to usual addition and scalar multiplication.
- ☐ The vector space R^5 R5(with respect to usual additional and scalar multiplication) contains two disjoint subspaces.
- ☐ There exist three linearly dependent vectors in R^3 R3 such that none of the three is a multiple of another.
- ☐ Union of two linearly independent sets need not be linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
There exist three linearly dependent vectors in R^3 R3 such that none of the three is a multiple of another.
Union of two linearly independent sets need not be linearly independent.
2 points

If S_1 S1 is a maximal linear independent set and S_2 S2 is a minimal spanning set of a vector space V V, then which of the following option(s) is (are) true?

☐ For any $v \in S_1$ v ∈ S1, $S_1 \setminus \{v\}$ S1 \ {v} is a linearly independent.

☐ For any $v \in S_2$ v ∈ S2, $S_2 \setminus \{v\}$ S2 \ {v} is a spanning set of V V.

☐ For any $v \in V \setminus S_1$ v ∈ V \ S1, $S_1 \cup \{v\}$ S1 ∪ {v} is a linearly dependent.

☐ For any $v \in V$ v ∈ V, $S_2 \cup \{v\}$ S2 ∪ {v} is a spanning set of V V.

No, the answer is incorrect.
Score: 0
Accepted Answers:
For any $v \in S_1$ v ∈ S1, $S_1 \setminus \{v\}$ S1 \ {v} is a linearly independent.
For any $v \in V \setminus S_1$ v ∈ V \ S1, $S_1 \cup \{v\}$ S1 ∪ {v} is a linearly dependent.
For any $v \in V$ v ∈ V, $S_2 \cup \{v\}$ S2 ∪ {v} is a spanning set of V V.

1 point

Consider the following statements:

Statement 1:Statement 1: $0(0,0) + 0(1,1) = (0,0)$0(0, 0) + 0(1, 1) = (0, 0) implies that the set $\{(0,0),(1,1)\}${(0, 0), (1, 1)} is a linearly independent subset of R^2 R2.
Statement 2:Statement 2: $2(1,0) + 2(0,1) = (2,2)$2(1, 0) + 2(0, 1) = (2, 2) implies that the set $\{(1,0),(0,1),(2,2)\}${(1, 0), (0, 1), (2, 2)} is a linearly dependent subset of R^2 R2.

Which of the following options is correct?
○ Both the statements are true.
○ Statement 1 is true, but Statement 2 is false.
○ Statement 2 is true, but Statement 1 is false.
○ Both the statements are false.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Statement 2 is true, but Statement 1 is false.

Three people denoted by P_1, P_2, P_3 P1, P2, P3 intend to buy some rolls, buns and cakes from a shop. They pay ₹ x_1 x1, ₹ x_2 x2, ₹ x_3 x3 per unit for rolls, buns and cakes, respectively. P_1 P1 bought 11 unit of rolls, 22 units of buns, and 33 units of cakes, P_2 P2 bought 22 units of rolls, 44 units of buns, and 55 units of cakes and P_3 P3 bought 11 unit of rolls, 33 units of buns, and 44 units of cakes. The total amount spent by P_1, P_2, P_3 P1, P2, P3 are ₹ 1616, ₹ 2929, ₹ 2121, respectively. Answer the following questions using this given data.

Represent the system of linear equations above in terms of its matrix form $Ax = b$Ax = b, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1 x2 x3], and $b = \begin{bmatrix} 16 \\ 29 \\ 21 \end{bmatrix}$b = [16 29 21]. Find the $(2,2)$(2, 2)-th cofactor of AA?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

If $x = [\begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}]$x = $\begin{bmatrix} \\ \text{x1x2x3} \end{bmatrix}$ is the solution of $Ax = b$Ax = b, then the value of $x_1 + x_2 + x_3$ x1 + x2 + x3 is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 8
2 points

Let ($M_{m \times n}(R)$)(Mm×n(R)) denote the set of all ($m \times n$)(m × n) matrices with real entries and let (D)(D) be a non-zero matrix such that (D)(D) is the product of the matrices ($A \in M_{7 \times 12}(R), B \in M_{12 \times 3}(R)$)(A ∈ M7×12(R), B ∈ M12×3(R)) and ($C \in M_{3 \times 5}(R)$)(C ∈ M3×5(R)).

Find the maximum possible rank of (D)(D)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Find the minimum possible rank of (D)(D)